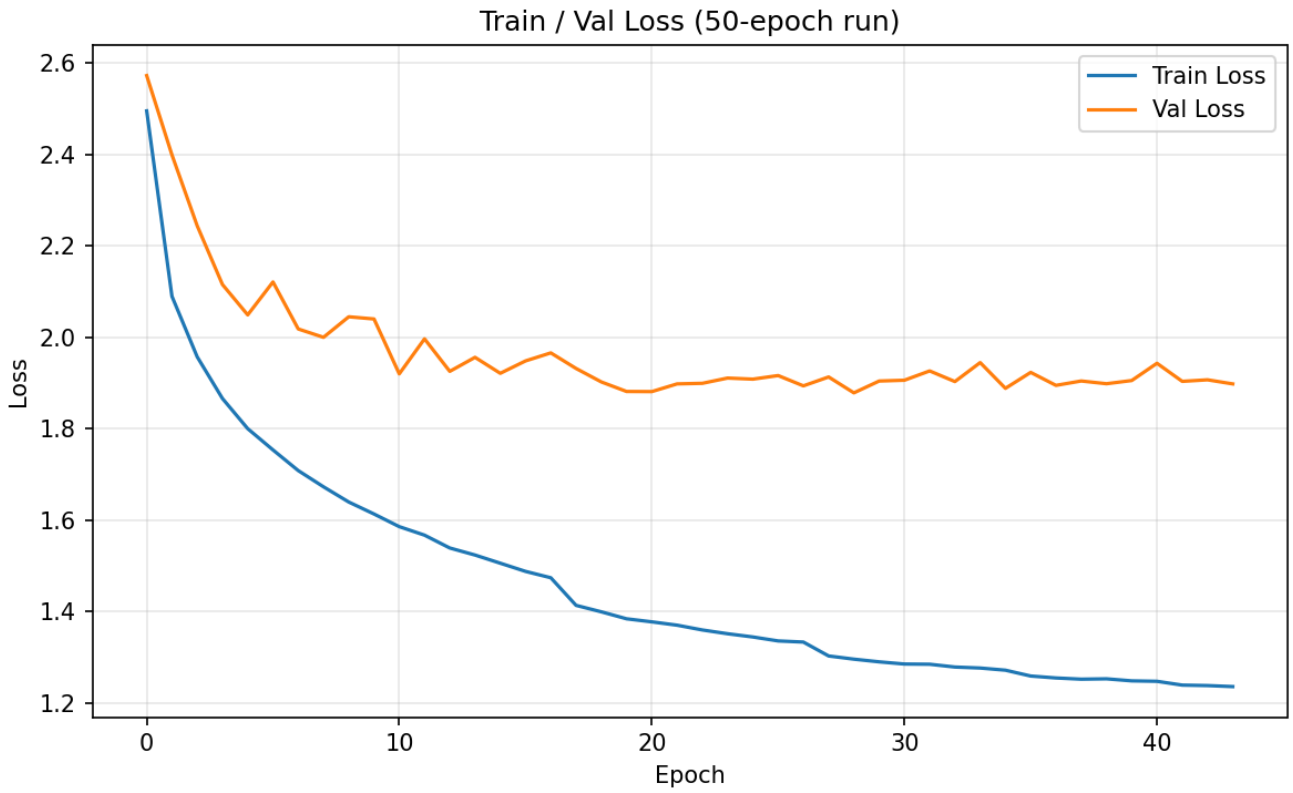
Model and training configuration:

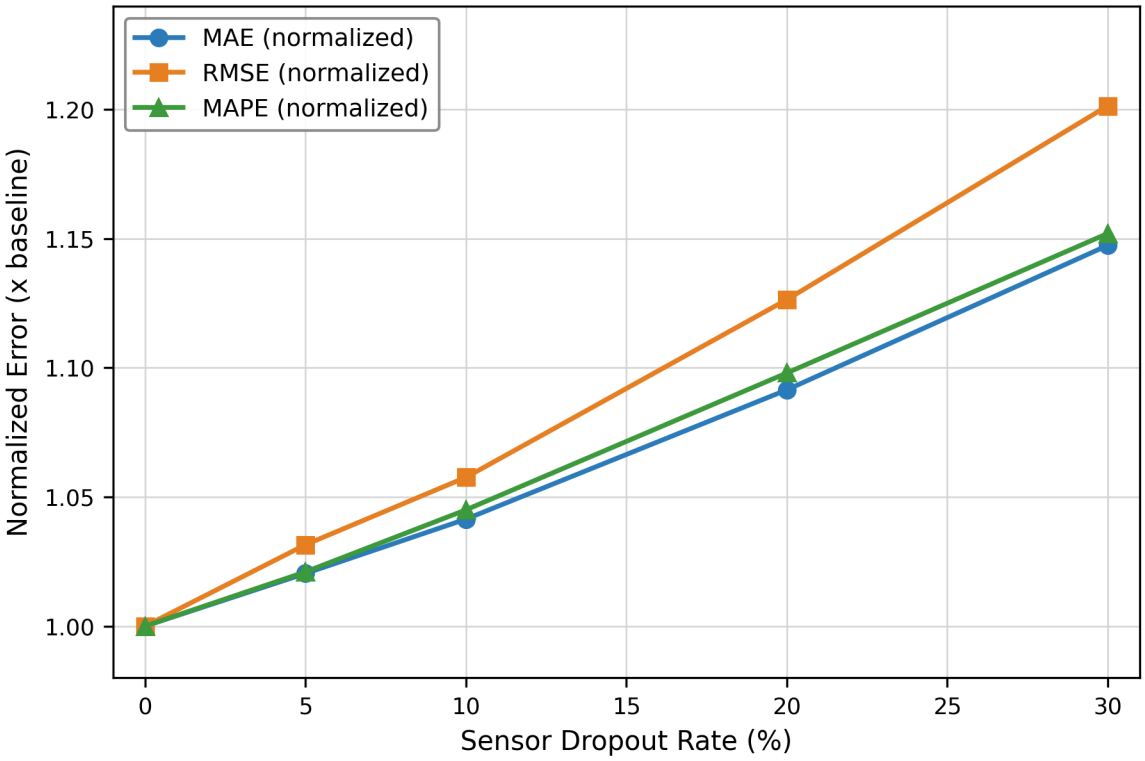
Component	Specification
Model Architecture	
Dataset	METR-LA
Number of Sensors	207
Input Window	12 time steps (60 minutes, 5-min intervals)
Prediction Horizon	12 time steps (60 minutes ahead)
Spatio-Temporal Blocks	4 stacked graph/spatio-temporal layers
Hidden Feature Dimension	64
Diffusion Steps	2-hop propagation
Graph Attention Heads	4
Temporal Convolution	Kernel size 2; dilations {1, 2, 4}
BiLSTM Hidden Size	64
Dropout Rate	0.15
Total Parameters	≈270K (270,340)
Training Setup	
Optimizer	AdamW (learning rate = 10^{-3}, weight decay = 10^{-5})
Scheduler	ReduceLROnPlateau (factor = 0.5, patience = 5)
Epochs / Batch Size	50 / 8
Loss Function	Uncertainty-aware loss: MSE + NLL
Gradient Clipping	max_norm = 1.0
Early Stopping	Patience = 15 epochs
Uncertainty Estimation	
MC Dropout	Enabled at inference
MC Samples	10 stochastic forward passes
Aleatoric Uncertainty	Learned through the variance prediction head
Epistemic Uncertainty	Estimated from variance across MC forward passes

Performance analysis on METR-LA test set:

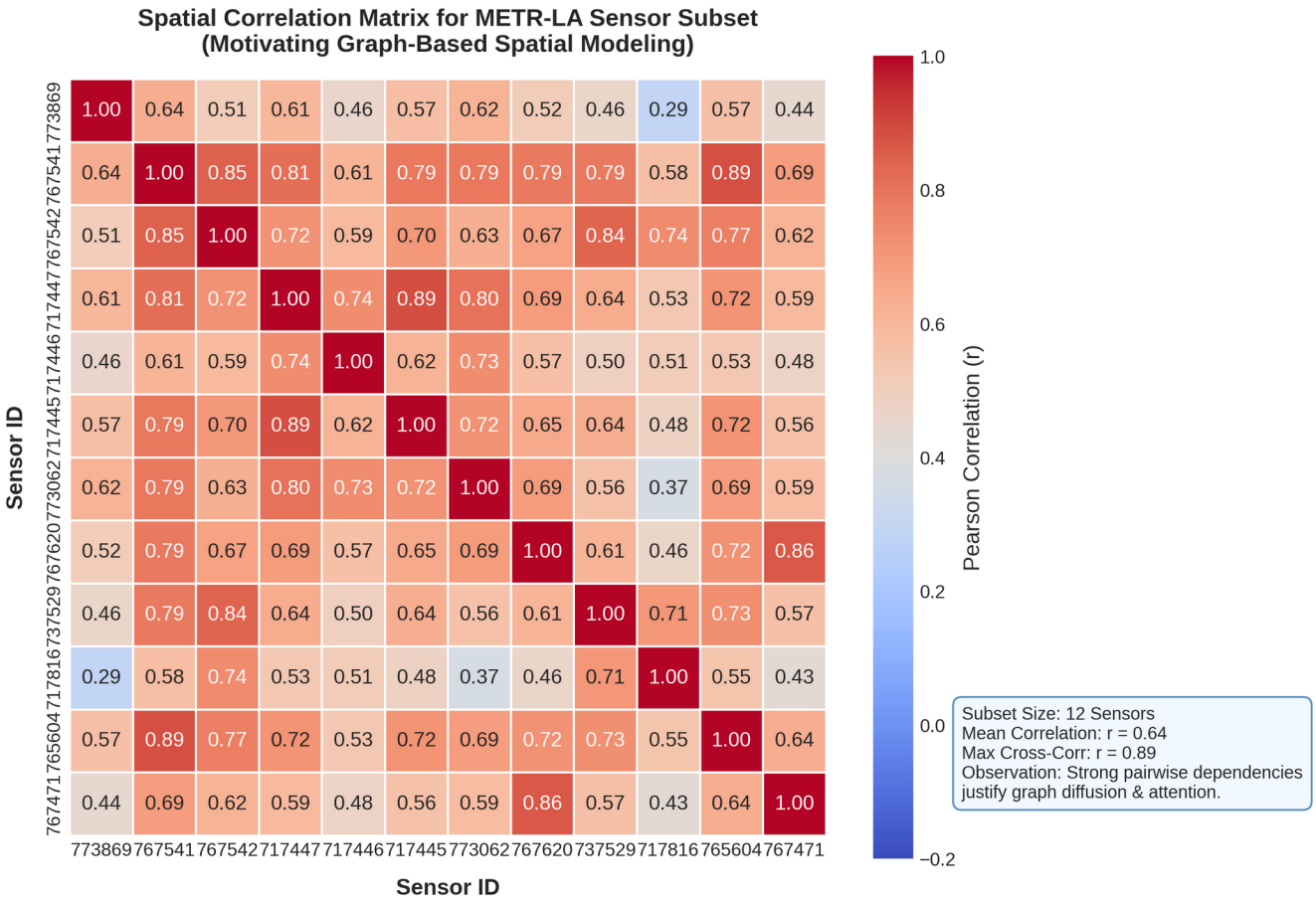
Metric	Value
MAE	0.6099
RMSE	1.3172
R2 Score	0.6620
Pearson Correlation	0.8137
Mean Aleatoric Uncertainty	1.2662
Mean Epistemic Uncertainty	0.2774
Mean Total Uncertainty	1.5437



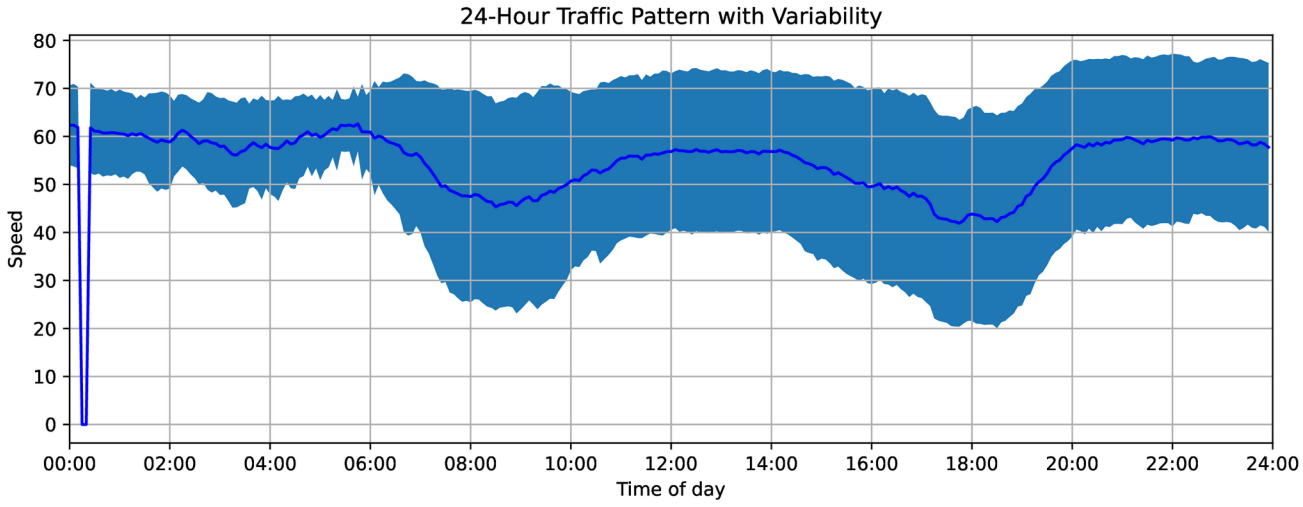
Training and validation loss curves over 50epochs, demonstrating smooth convergence and stable gen-eralization behavior on the METR-LA dataset.



Robustness of UA-GNN under sensor dropout on the METR-LA dataset



Correlation matrix of a sensor subset, motivating graph-based spatial modeling.



Representative 24-hour traffic speed profile with predictive uncertainty band. Wider intervals correspond to peak-hour variability, while narrower bands indicate stable traffic conditions.

Ablation Study: Component contribution over-all Performance on METR-LA Dataset:

Model Variant	MAE	RMSE	MAE Degradation (%)
Full Model (all components)	0.6099	1.3172	Baseline
w/o Diffusion Conv (GCN only)	1.9693	2.8884	+222.89%
w/o BiLSTM (Temporal Conv only)	1.8852	3.0321	+209.10%
w/o Graph Attention	0.8106	1.4782	+32.91%
w/o MC Dropout (Deterministic)	0.6133	1.3489	+0.56%

Computational Efficiency Comparison on METR-LA:

Model	Inference Time (ms)	Memory (MB)	Parameters (K)	Notes / Source
Proposed Model	**59.3 ± 0.7**	**358** (train buffer) / **99.8** (batch=32) / **95.4 ± 1.3** (batch=8 eval)	**270.3** (core) / **303.8** (w/ embeddings)	Empirically profiled on GPU (exact weights: 270,340)
DCRNN	150.9 ± 3.4	1,417	1,200.0	207/325 graph node scaling from published baseline
Graph WaveNet	86.8 ± 2.1	803	700.0	207/325 graph node scaling from published baseline
Bayesian LSTM	113.4 ± 2.8	1,057	900.0	207/325 node scaling from published baseline

UA-GNN FINAL EVALUATION SUMMARY (METR-LA DATASET)

1. OVERALL METRICS (Standardized / Normalized Latent Space):
   - MAE: 0.609916
   - RMSE: 1.317238
   - R^2 Score: 0.661991
   - Pearson Correlation: 0.813655

- Mean Aleatoric Uncertainty: 1.266244
- Mean Epistemic Uncertainty: 0.277434
- Mean Total Uncertainty: 1.543679

2. PHYSICAL SPEED METRICS (Miles Per Hour, mph):
- Overall MAE: 3.4642 mph
- Overall RMSE: 6.5746 mph
- Overall MAPE: 9.65 %

3. HORIZON-SPECIFIC METRICS (Physical mph):
- 15 min (step 3): MAE = 3.0592 mph, RMSE = 5.6100 mph
- 30 min (step 6): MAE = 3.4616 mph, RMSE = 6.5416 mph
- 45 min (step 9): MAE = 3.7671 mph, RMSE = 7.2030 mph
- 60 min (step 12): MAE = 4.0401 mph, RMSE = 7.7800 mph

4. CALIBRATION:
- Expected Calibration Error (ECE): 0.1614